

核心文理-別冊解答

大島 遙斗

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1 講義問題-解答

1.1 差分の基礎

1.1

▶解答◀

$$(1) \Delta f(k) = k + 1 - k = 1$$

$$(2) \Delta f(k) = (k + 1)^2 - k^2 = 2k + 1$$

$$(3) \Delta f(k) = (k + 1)^3 - k^3 = 3k^2 + 3k + 1$$

$$(4) \Delta f(k) = (k + 4)(k + 3)(k + 2)(k + 1) - k(k + 1)(k + 2)(k + 3) = 4(k + 1)(k + 2)(k + 3)$$

$$(5) \Delta f(k) = 2^{k+1} - 2^k = 2^k$$

$$(6) \Delta f(k) = 3^{k+1} \cdot (k + 1)^2 - 3^k \cdot k^2 = 3^k(2k^2 + 6k + 3)$$

$$(7) \Delta f(k) = \frac{1}{(k + 1)!} - \frac{1}{k!} = -\frac{k}{(k + 1)!}$$

$$(8) \Delta f(k) = 0$$

$$(9) \Delta f(k) = 0$$

1.2

▶解答◀

$$(1) A_n = \sum_{k=1}^n 1 = \left[k \right]_1^{n+1} = n$$

$$(2) B_n = \sum_{k=1}^n k = \sum_{k=1}^n \Delta \left(\frac{k(k-1)}{2} \right) = \left[\frac{k(k-1)}{2} \right]_1^{n+1} = \frac{n(n+1)}{2}$$

$$(3) C_n = \sum_{k=1}^n k^2 \text{ を求める.}$$

$k^2 = k(k-1) + k$ と変形できるので,

$$\begin{aligned} C_n &= \sum_{k=1}^n \{k(k-1) + k\} \\ &= \left[\frac{(k-2)(k-1)k}{3} + \frac{k(k-1)}{2} \right]_1^{n+1} \\ &= \frac{1}{3}(n-1)n(n+1) + \frac{1}{2}n(n+1) \\ &= \frac{1}{6}n(n+1)(2n-2+3) \\ &= \frac{1}{6}n(n+1)(2n+1) \end{aligned}$$

(4) $D_n = \sum_{k=1}^n k^3$ を求める.

$k^3 = k(k-1)(k+1) + k$ と変形できるので,

$$\begin{aligned}
 D_n &= \sum_{k=1}^n \{k(k-1)(k+1) + k\} \\
 &= \left[\frac{(k-2)(k-1)k(k+1)}{4} + \frac{(k-1)k}{2} \right]_1^{n+1} \\
 &= \frac{(n-1)n(n+1)(n+2)}{4} + \frac{n(n+1)}{2} \\
 &= \frac{1}{4} \{ (n-1)n(n+1)(n+2) + 2n(n+1) \} \\
 &= \frac{1}{4} \{ n(n+1)(n^2 + n - 2 + 2) \} \\
 &= \frac{1}{4} n(n+1)n(n+1) \\
 &= \frac{1}{4} n^2(n+1)^2
 \end{aligned}$$

(5)

$$\begin{aligned}
 E_n &= \sum_{k=0}^{n^2+1} k(k+1)(k+2)(k+3) \\
 &= \left[\frac{1}{5} (k-1)k(k+1)(k+2)(k+3) \right]_0^{n^2+1} \\
 &= \frac{1}{5} (n^2+1)(n^2+2)(n^2+3)(n^2+4)(n^2+5)
 \end{aligned}$$

(6) $F_n = \sum_{k=-3}^{n-1} (k+2)(k+3)(k+4)(k+4)$ を求める.

まず, k の式を連続整数積の形に変形する.

$$\begin{aligned}
 (k+2)(k+3)(k+4)(k+4) &= (k+2)(k+3)(k+4)(k+1+3) \\
 &= (k+1)(k+2)(k+3)(k+4) + 3(k+2)(k+3)(k+4)
 \end{aligned}$$

このことより,

$$\begin{aligned}
 F_n &= \left[\frac{1}{5} k(k+1)(k+2)(k+3)(k+4) + 3 \cdot \frac{1}{4} (k+1)(k+2)(k+3)(k+4) \right]_{-3}^{n-1} \\
 &= \frac{1}{5} (n-1)n(n+1)(n+2)(n+3) + \frac{3}{4} n(n+1)(n+2)(n+3) \\
 &= \frac{1}{20} (n+1)(n+2)(n+3)(n+4)(4n+15) \\
 &= \frac{1}{20} (n+1)(n+2)(n+3)(n+4)(4n+15)
 \end{aligned}$$

(7)

$$\begin{aligned} G_n &= \sum_{k=1}^n 2^k \\ &= \left[\frac{2^k}{2-1} \right]_1^{n+1} \\ &= 2^{n+1} - 2 \\ &= 2 \cdot (2^n - 1) \end{aligned}$$

(8) $H_n = \sum_{k=1}^n (k+2) \cdot 3^k$ を求める.

今, $f(k) = (k+2) \cdot 3^k$, $F(k) = (pk^2 + qk + r) \cdot 3^k$ (p, r, q は定数) とおいて, $\Delta F(k) = f(k)$ を満たすように定数 p, q, r を定める.

$$\begin{aligned} \Delta F(k) &= \{p(k^2 + 2k + 1) + q(k+1) + r\} \cdot 3^{k+1} - (pk^2 + qk + r) \cdot 3^k \\ &= 3^k \cdot \{2pk^2 + 2(3p+q)k + 3p + 2q + 2r\} \end{aligned}$$

$f(k)$ とそれぞれ係数比較して,

$$\begin{cases} 2p = 0 \\ 6p + 2q = 1 \\ 3p + 2q + 2r = 2 \end{cases} \iff \begin{cases} p = 0 \\ q = \frac{1}{2} \\ r = \frac{1}{2} \end{cases}$$

$$\therefore F(k) = \frac{3^k}{2}(k+1)$$

よって,

$$\begin{aligned} H_n &= \sum_{k=1}^n F(k) \\ &= \left[\frac{3^k}{2}(k+1) \right]_1^{n+1} \\ &= \left(\frac{n}{2} + 1 \right) \cdot 3^{n+1} - 3 \end{aligned}$$

(9) $I_n = \sum_{k=1}^n k^4$ を求める.

まず, k^4 を連続整数積の形に変形する.

$$\begin{aligned} k^4 &= k^2(k-1)(k+1) + k^2 = k(k-1)k(k+1) + k^2 = (k-2+2)(k-1)k(k+1) + k^2 \\ &= (k-2)(k-1)k(k+1) + 2(k-1)k(k+1) + k^2 \end{aligned}$$

よって,

$$\begin{aligned} I_n &= \left[\frac{1}{5}(k-3)(k-2)(k-1)k(k+1) + \frac{2}{4}(k-2)(k-1)k(k+1) \right]_1^{n+1} + \frac{1}{6}n(2n+1)(n+1) \\ &= \frac{1}{5}(n-2)(n-1)n(n+1)(n+2) + \frac{1}{2}(n-1)n(n+1)(n+2) + \frac{1}{6}n(2n+1)(n+1) \\ &= \frac{1}{30}n(n+1)\{6(n-2)(n-1)(n+2) + 15(n-1)(n+2) + 5(2n+1)\} \\ &= \frac{1}{30}[3(n-1)(n+2)\{2(n-2) + 5\} + 5(2n+1)] \\ &= \frac{1}{30}\{3(n-1)(n+2)(2n+1) + 5(2n+1)\} \\ &= \frac{1}{30}n(n+1)[(2n+1)\{3n^2 + 3n - 6 + 5\}] \\ &= \frac{1}{30}n(n+1)(2n+1)(3n^2 + 3n - 1) \end{aligned}$$

1.3

▶解答◀

(1)

$$\begin{aligned}\sum_{k=1}^n \{ (k+1)^7 - k^7 \} &= \sum_{k=1}^n \Delta(k^7) \\ &= \left[k^7 \right]_1^{n+1} \\ &= \mathbf{k^7 - 1}\end{aligned}$$

(2)

$$\begin{aligned}\sum_{k=1}^n \frac{1}{\sqrt{k+1} + \sqrt{k}} &= \sum_{k=1}^n (\sqrt{k+1} - \sqrt{k}) \\ &= \left[\sqrt{k} \right]_1^{n+1} \\ &= \mathbf{\sqrt{n+1} - 1}\end{aligned}$$

(3)

$$\begin{aligned}\sum_{k=1}^n \frac{1}{k(k+1)} &= \sum_{k=1}^n \left(\frac{1}{k} - \frac{1}{k+1} \right) \quad (\text{部分分数分解}) \\ &= \left[\frac{1}{k} \right]_1^{n+1} \\ &= \mathbf{\frac{1}{n+1} - 1}\end{aligned}$$

1.2 差分の応用

2.1

▶解答◀ (1)

$$\begin{aligned} A_n &= \sum_{k=-n}^n 3^k = \frac{3^n \cdot 2 - 3^{-n}}{3 - 1} \left(\frac{(\text{最後の次}) - (\text{最初})}{(\text{公比}) - 1} \right) \\ &= \frac{3^n}{2} \cdot \frac{8}{3} \\ &= 4 \cdot 3^{n-1} \end{aligned}$$

$$(2) B_n = \sum_{k=1}^n k \cdot 2^{-k}$$

$f(k) = k \cdot 2^{-k}$, $F(k) = (pk^2 + qk + r)$ (p, q, r は定数) において, $\Delta F(k) = f(k)$ を満たすように p, q, r を定める.

$$\begin{aligned} \Delta F(k) &= 2^{-(k+1)} \{ p(k+1)^2 + q(k+1) + r \} - 2^{-k} \{ pk^2 + qk + r \} \\ &= 2^{-k} \left\{ -\frac{1}{2} pk^2 + \left(p - \frac{q}{2} \right) k + \frac{p}{2} + \frac{q}{2} - \frac{r}{2} \right\} \end{aligned}$$

$f(k)$ とそれぞれ係数比較して,

$$\begin{cases} p = 0 \\ -\frac{q}{2} = 1 \\ -1 - \frac{r}{2} = 0 \end{cases} \iff \begin{cases} p = 0 \\ q = -2 \\ r = -2 \end{cases}$$

$$\therefore F(k) = -(k+1) \cdot 2^{-k+1}$$

$$\begin{aligned} B_n &= \left[-(k+1) \cdot 2^{-k+1} \right]_1^{n+1} \\ &= -\left(\frac{n+2}{2^n} - 2 \right) \\ &= 2 - \frac{n+2}{2^n} \end{aligned}$$

(3) $\sum_{k=1}^n k^2 \cdot 3^k$ を求める.

$f(k) = k^2 \cdot 3^k$, $F(k) = (pk^3 + qk^2 + rk + s) \cdot 3^k$ (p, q, r, s は定数) において, $\Delta F(k) = f(k)$ を満

たすように p, q, r, s を定める.

$$\begin{aligned}\Delta F(k) &= (p(k+1)^3 + q(k+1)^2 + r(k+1) + s) \cdot 3^{k+1} - (pk^3 + qk^2 + rk + s) \cdot 3^k \\ &= 3^k [3\{p(k^3 + 2k^2 + 2k + 1) + q(k^2 + 2k + 1) + r(k+1) + s\} - (pk^3 + qk^2 + rk + s)]\end{aligned}$$

以下, $[\]$ の中だけ計算する.

$$\begin{aligned}[\] &= 2pk^3 + 6pk^2 + 5qk^2 + 6pk + 6qk + 2rk + 3p + 3q + 3r + 2s \\ &= 2pk^3 + (6p + 5q)k + (6p + 6q + 2r)k + 3p + 3q + 3r + 2s\end{aligned}$$

$f(k)$ それぞれ係数比較して,

$$\begin{cases} p = 0 \\ q = \frac{1}{5} \\ \frac{6}{5} + 2r = 0 \\ \frac{3}{5} + 3r + 2s = 0 \end{cases} \iff \begin{cases} p = 0 \\ q = \frac{1}{5} \\ r = -\frac{3}{5} \\ s = \frac{3}{5} \end{cases}$$

$$\therefore F(k) = \left(\frac{1}{5}k^2 - \frac{3}{5}k + \frac{3}{5} \right) \cdot 3^k$$

$$\begin{aligned}\therefore C_n &= \left[\frac{1}{5}k^2 - \frac{3}{5}k + \frac{3}{5} \cdot 3^k \right]_1^{n+1} \\ &= \frac{1}{5} \{ ((n+1)^2 - 3(n+1) + 3) \cdot 3^{n+1} - 3 \} \\ &= \frac{1}{5} \{ (n^2 - n + 1)3^{n+1} - 3 \} \\ &= \frac{3}{5} \{ (n^2 - n + 1) \cdot 3^n - 1 \}\end{aligned}$$

2.2

考え方

見た目でも単調減少な数列であるので、 $n = 1$ で最大になることは予想できます。あとはそれを示すだけです。数列の単調性を示す方法は次の2通りです。数学 III でも頻出なので勉強しておきましょう。

数列の単調性の証明

a_n をある数列とする。 $n \in \mathbb{N}$

- $a_{n+1} - a_n < 0 \iff a_{n+1} < a_n \iff a_n$ は単調減少数列.
- $a_{n+1} - a_n > 0 \iff a_{n+1} > a_n \iff a_n$ は単調増加数列.
- $\frac{a_{n+1}}{a_n} < 1 \iff a_{n+1} < a_n \iff a_n$ は単調減少数列.
- $\frac{a_{n+1}}{a_n} > 1 \iff a_{n+1} > a_n \iff a_n$ は単調増加数列.

▶解答◀ $f(n) = n\left(\frac{20}{21}\right)^n$ を最大にする自然数 n を求める.

$$\begin{aligned}\Delta f(n) &= (n+1)\left(\frac{20}{21}\right)^{n+1} - n\left(\frac{20}{21}\right)^n \\ &= \frac{20}{21} \left\{ \left(\frac{20}{21}\right)^n (n+1) - n \right\} < 0\end{aligned}$$

$\therefore f(n)$ は全ての自然数 n に対して、単調減少な数列なので、 $f(n)$ を最大にする n は $n = 1$.

2.3

考え方 2.2 と方針は全く同じです. $2^n - n^2$ が単調増加なことを示します.

▶解答◀ 全ての4以上の自然数 n に対し, $2^n \geq n^2$ を示す. $f(n) = 2^n - n^2$ とおく.

$$\begin{aligned}\Delta f(n) &= 2^{n+1} - (n+1)^2 - (2^n - n^2) = 2^n - 2n - 1 > 0 \cdots \textcircled{1} \quad (\text{《注》を参照せよ.}) \\ &\iff 2^n > 2n + 1\end{aligned}$$

よって, $f(n)$ は単調増加な数列である.

実際に,

$$f(4) = 0$$

$$f(5) = 7$$

$$f(6) = 28$$

⋮

すなわち, $f(4) < f(5) < f(6) < \cdots$

よって, $f(n) \geq 0 \iff 2^n \geq n^2$ ■

《注》① 式で > 0 と判断して良い理由. $\rightarrow$ 発散スピードと収束スピードが根拠.

📖 収束スピードと発散スピード

一般に, 関数もしくは数列に関して, 次のことが言える.

$$\cdots \ll \log x \ll \cdots \ll \sqrt{x} \ll x \ll x^2 \ll \cdots \ll 2^x \ll e^x \ll 3^x \ll \cdots \ll x^x \ll \cdots$$

従って, ①の式は(指数関数)-(一次関数)という形なので十分大きい n に対して, 指数関数の方が圧倒的に影響が大きい! ので, > 0 と言って良いのです.

別解(数学 III 利用)

命題は, 実数 x に対して, $2^x \geq x^2$ を示すことと同値である. $f(x) = 2^x - x^2$ とおくと,

$$f'(x) = \{e^{\log 2^x} - x^2\}' = \{e^{x \log 2} - x^2\}' = \log 2 \cdot 2^x - 2x$$

$$f'(\log 2) = 2 \log 2 - 2 \log 2 = 0$$

よって, 次の増減表を得る.

x	0	⋯	$\log 2$	⋯	4		∞
$f'(x)$			0	+	+	+	
$f(x)$			-2	↗	0	↗	∞

よって, 上表から4以上のすべての自然数に対して, $f(x) \geq 0$ なので, $2^n \geq n^2$ ■

1.3 漸化式

ほげ

1.4 二項定理

ほげ

1.5 場合の数が一番大切なこと

ほげ

1.6 種々の数え上げ

ほげ

2 各章の問題演習-解答

2.1 1章：差分の基礎

1.

▶解答◀ (1)

$$\begin{aligned} A_n &= \sum_{k=2}^n \frac{1}{k(k-1)} = \sum_{k=2}^n \left(\frac{1}{k-1} - \frac{1}{k} \right) = \left[\frac{1}{k-1} \right]_2^{n+1} \\ &= \frac{1}{n} - 1 \end{aligned}$$

(2) $B_n = \sum_{k=1}^n \frac{1}{k(k+1)(k+2)}$ を求める.

$$\frac{1}{k(k+1)(k+2)} = \frac{1}{2} \left\{ \frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} \right\} \text{ と変形できるので,}$$

$$\begin{aligned} \therefore B_n &= -\frac{1}{2} \sum_{k=1}^n \left\{ \frac{1}{k(k+1)} - \frac{1}{(k+1)(k+2)} \right\} = \left[\frac{1}{k(k+1)} \right]_1^{n+1} = \frac{n^2 + 3n + 1}{4(n+2)(n+1)} \\ (3) \end{aligned}$$

$$C_n = \sum_{k=1}^n k(k-1) = \left[\frac{1}{3}(k-1)k(k+1) \right]_1^{n+1} = \frac{1}{3}n(n+1)(n+2)$$

(4)

$$\begin{aligned} D_n &= \sum_{k=-3}^n (k+2)(k+3)(k+4) \\ &= \sum_{k=-3}^n (k+2)(k+3)(k+4) \\ &= \left[\frac{1}{4}(k+1)(k+2)(k+3)(k+4) \right]_{-3}^{n+1} \\ &= \frac{1}{4}(n+2)(n+3)(n+4)(n+5) \end{aligned}$$

(5)

$$\begin{aligned} E_n &= \sum_{k=0}^n (n-k)(n-(k-1))(n-(k-2)) \\ &= \sum_{k=0}^n \Delta \left(\frac{1}{4}(n-(k-3))(n-(k-2))(n-(k-1))(n-k) \right) \\ &= \left[\frac{1}{4}(n-(k-3))(n-(k-2))(n-(k-1))(n-k) \right]_0^{n+1} \\ &= \frac{1}{4} \cdot 2 \cdot 1 \cdot 0 \cdot 1 - \left\{ \frac{1}{4}(n+3)(n+2)(n+1)n \right\} \\ &= -\frac{1}{4}n(n+1)(n+2)(n+3) \end{aligned}$$

2.2 2 章：差分の応用